SAPONIFICATION OF AN ESTER (Ethyl Acetate)

I. Objective

Study the kinetics of a saponification reaction which takes place between ethyl acetate $CH_3COOC_2H_5$ and sodium hydroxide NaOH.

II. Principle

The addition of a sodium hydroxide solution to a solution of ethyl acetate $CH_3COOC_2H_5$ results in the hydrolysis reaction:

$$CH_3COOC_2H_5 + OH^- \longrightarrow CH_3COO^- + C_2H_5OH$$

On s'intéresse aux premiers instants de la réaction. Dans ces conditions, les concentrations en ions acétate et en éthanol restent suffisamment faibles pour que la réaction inverse puisse être négligée.

III. Materials and Products

Materials

The equipment necessary to carry out the study: - Burettes. - A pipette. - An Erlenmeyer flask.

- Beakers - A magnetic stirrer. - A magnetic bar. -A wash bottle. - Heating plate.

Solutions

The solutions necessary to carry out the dosage:

- Ethyl acetate $CH_3COOC_2H_5$ de 5.10^{-2} M.

- Sodium hydroxide NaOH de 5.10^{-2} M.

- Hydrochloric acid HCL à 10^{-2} M.

- Color indicator: phenolphthalein.

IV. Operating mode

At time t = 0, mix a volume V_1 = 50 ml of ester (ethyl acetate) with a concentration $C_1 = 5.10^{-2}$ M and a volume V_2 = 50 ml of Sodium Hydroxide with a concentration $C_2 = 5.10^{-2}$ M with stirring (To carry out this stirring, we use a magnetic stirrer and a magnetic bar).

At time t we take 10 mL of this mixture containing x mol of OH^- ions, at t = 0. This quantity decreases over time due to the consumption of this species by the saponification reaction. Return dosages make it possible to determine the quantity of soda consumed.

At regular time intervals, a volume of 10 cm^3 of the reaction mixture is taken using a pipette.

This test portion is quickly emptied into an Erlenmeyer flask and the remaining ions are measured with a solution of hydrochloric acid in the presence of phenolphthalene.

Dosage of excess acid.

The V_{HCL} volumes of 10^{-2} M necessary for the dosages of the excesses of the mixture present in the Erlenmeyer flasks are reported in the table: The concentration Ct of sodium hydroxide in the reaction mixture at time t is obtained by equalizing the quantities of acid and base present at the equivalent point of the dosage.

By repeating this process for times 4, 9, 8, 15…..and 47 min, we complete the table:

Erlenmeyer n°	1	2	3	4	5	6	7	8
t (min)	0	4	9	15	24	30	40	47
V_{HCL} (mL)								
$C_{C_2H_5OH}$ (mol.L^{-1})								

V. Questions:

1- Write a personal report relating to this practical work.

2- Write the chemical reaction.

3- Complete by calculating for each sample the concentration of the ethanol formed.

4- Draw the curve C_2H_5OH = f(t)

5- At what time is the rate of formation of ethanol greatest? Justify.

6- Conclusion

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